

7.1.7 Multilayer Artificial Neural Network *

After Rosenblatt perceptron was developed in the 1950s, there was a lack of interest in neural networks until 1986, when Dr. Hinton and his colleagues developed the backpropagation algorithm to train a multilayer neural network.

Today it is a hot topic with many leading firms like Google, Facebook, and Microsoft which invest heavily in applications using deep neural networks. A fully connected multilayer neural network is called a *multilayer perceptron (MLP)*.

Backpropagation *

Backpropagation is a common method for training a neural network. Here, how it works is explained with a concrete example. Consider an example with two inputs, two hidden neurons, and two output neurons. Additionally, the hidden and output neurons will include a bias.

Here's the basic structure to implement X-OR gate:

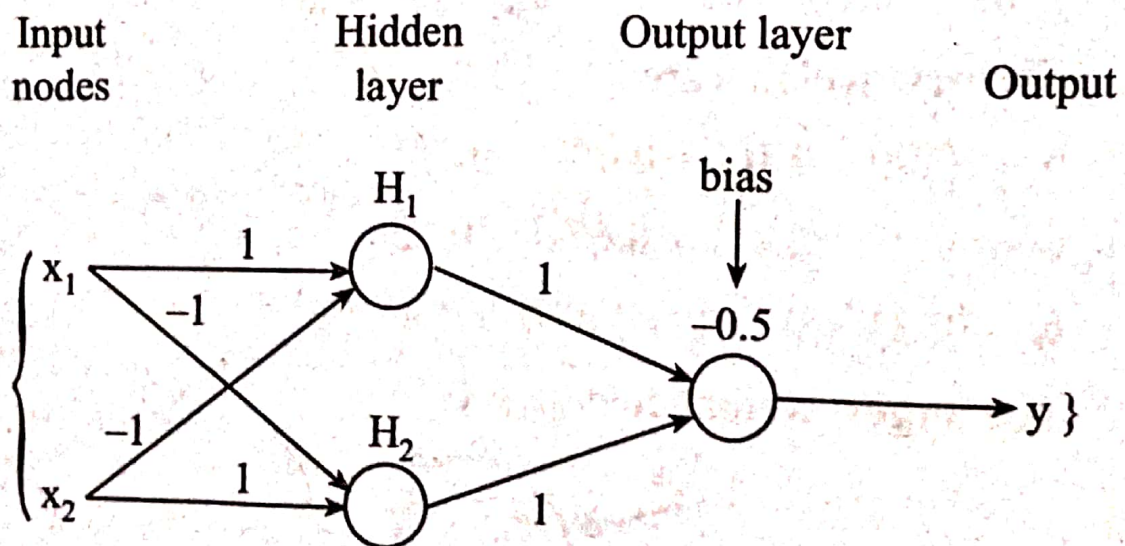


Figure 7.3: Feed forward neural network for XOR gate

Table: Working of an feed forward neural network for XOR gate

Input		Output of hidden nodes		Output node	$x_1 \text{ XOR } x_2$
x_1	x_2	H_1	H_2		
0	0	0	0	$-0.5 \rightarrow 0$	0
0	1	$-1 \rightarrow 0$	1	$0.5 \rightarrow 1$	1
1	0	1	$-1 \rightarrow 0$	$0.5 \rightarrow 1$	1
1	1	0	0	$-0.5 \rightarrow 0$	0

Backpropagation Algorithm

Step 1: Initialize all weights and bias randomly.

Step 2: Consider training data set and corresponding input as $I_i = O_i$ and $T = Y_i$.

Step 3: Do forward pass to calculate hidden layer values

$$I_j = \sum(w_{ij} \times O_i) + b_j$$

If we take sigmoid function as activation function then

$$O_j = \frac{1}{1 + e^{-I_j}}$$

Step 4: Calculate for output layer,

$$I_k = \sum(w_{jk} \times O_j) + b_k$$

$$O_k = \frac{1}{1 + e^{-I_k}}$$

Step 5: Calculate errors:

$$\text{error in output layer, } E_k = O_k(1 - O_k)(T - O_k)$$

$$\text{error in hidden layer, } E_j = O_j(1 - O_j) \sum(E_k \times w_{jk})$$

Step 6: Hidden layer's weight and bias are updated using following formula:

$$\Delta w_{ij} = \alpha \times E_j \times O_i$$

$$w_{ij} = w_{ij} + \Delta w_{jk}$$

$$\Delta b_j = \alpha \times E_j$$

$$b_j = b_j + \Delta b_j$$

Output layer weight and bias are updated using following formula:

$$\Delta w_{jk} = \alpha \times E_k \times O_j$$

$$W_{jk(\text{New})} = W_{jk(\text{old})} + \Delta w_{jk}$$

$$\Delta b_k = \alpha \times E_k$$

$$b_k = b_k + \Delta b_k$$

Step 7: Continue from steps (ii) to (vi) until all training sets values are satisfied.